

Background. Beyond ATP production, mitochondria act as central signaling platforms regulating cell death and inflammatory pathways such as NF- κ B and cGAS/STING, linking cellular bioenergetics to tissue-level inflammation. Alterations in mitochondrial respiration are increasingly recognized as upstream drivers of cytokine and chemokine production, particularly in immune cells whose functional state is tightly coupled to metabolic status in the central nervous system. The retina offers a uniquely sensitive system to study these processes: as a metabolically demanding neural tissue with a finely organized vascular network, it depends on tight regulation of inflammation and vascular growth. Disruption of this balance underlies blinding diseases such as diabetic retinopathy and age-related macular degeneration, in which inflammation is a key trigger of pathological angiogenesis. Myeloid cells are critical sensors of retinal stress and orchestrators of inflammatory and angiogenic responses, with distinct metabolic and mitochondrial features guiding their activation and functional states. How these metabolic features perpetuate inflammation in the central nervous system, however, remains unclear.

Methods: we employ a knockout of COX10 in CX3CR1⁺ cells to induce controlled Complex IV deficiency selectively in the immune compartment. Mitochondrial respiration is assessed by Seahorse extracellular flux analysis. The retinal phenotype is characterized *in vivo* using OCT, BluePeak autofluorescence imaging, and optomotor response, complemented by histology, immunofluorescence, single-cell transcriptomics, and metabolomics.

Results. Deletion of COX10 in CX3CR1⁺ cells reduce oxygen consumption, confirming cell-intrinsic Complex IV deficiency. *In vivo* retinal imaging at the timepoints examined shows preserved retinal thickness, autofluorescence, and visual acuity, defining a pre-symptomatic window in which to dissect molecular and cellular pathology preceding functional decline. Ongoing analyses focus on microglial activation, transcriptional reprogramming, and immune-endothelial crosstalk within this window.

Conclusions. This project establishes a tractable *in vivo* system to dissect how mitochondrial dysfunction in CX3CR1⁺ immune cells reshape inflammatory signaling and contributes to pathological retinal vascular growth, positioning immune-cell mitochondrial metabolism as a candidate upstream regulator of retinal inflammation and a potential therapeutic node.